



LABORATOIRE
JEAN KUNTZMANN
MATHÉMATIQUES APPLIQUÉES - INFORMATIQUE



PROBABILITY DISTRIBUTION FITTING AND SELECTION

FROM R TO JULIA

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OUTLINES

1 INTRODUCTION

2 LITERATURE REVIEW

- Probability distribution fitting
- How to do it in R? as of 2008

3 THE FITDISTRPLUS PROJECT

- Overview
- For what purpose, **fitdistrplus** is used

4 COMPARISON AND SELECTION OF DISTRIBUTION FITS

- Quantitative
- Graphics

5 EXTENSIONS AND INHERITANCE

- the mbbefd package
- the OneStep package

6 JULIA

7 CONCLUSION

The **fitdistrplus** project

- started in 2009: stable version 1.0-9 on CRAN (first release),
- extensively enhanced between 2009-2018: 17 versions on CRAN,
- published 2015: publication in JSS,
- currently, 2026: last stable version 1.2-6.

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Presented at

- [useR 2009](#) in Rennes, [useR 2011](#) in Warwick, [useR 2019](#) in Toulouse,
- [Rencontres R 2013](#) in Lyon, [Rencontres R 2018](#) in Rennes.

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fitdistrplus extends the `fitdistr()` function (of the MASS package).

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Assuming a sample $(X_i)_{1 \leq i \leq n} \stackrel{i.i.d.}{\sim} X$, the likelihood

$$\mathcal{L}(\theta, x_1, \dots, x_n) = \prod_{i=1}^n f_X(x_i, \theta),$$

where f_X is the mass probability/density function.

The Maximum Likelihood Estimator θ_{MLE} maximizes the likelihood $\mathcal{L}$ or $\log \mathcal{L}$.

$$\theta_{\text{MLE}} = \arg \max_{\theta \in \Theta} \mathcal{L}(\theta, x_1, \dots, x_n).$$

- Some closed-form expression, e.g., when f_X belongs to the exponential family.
- Generally, it requires a numerical optimization $\theta \mapsto \log \mathcal{L}(\theta, x_1, \dots, x_n)$.
- Initial guess is typically the moment matching or quantile matching estimator.

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`fitdistr()` allows to compute MLE for a selected set of distributions, not all of base R Distributions.

```
> library(MASS)
> set.seed(123)
> x <- rgamma(100, shape = 5, rate = 0.1)
> fitdistr(x, "gamma", list(shape = 1, rate = 0.1), lower = 0.001)
      shape      rate
6.48686551 0.13651012
(0.89438956) (0.01956818)
```

AT THE HEART OF `fitdistr()`

`fitdistr()` relies mostly on closed-form estimators.

```
fitdistr <- function(x, densfun, start, ...)  
{  
  myfn <- function(parm, ...) -sum(log(dens(parm, ...)))  
  mylogfn <- function(parm, ...) -sum(dens(parm, ..., log = TRUE))  
  ...  
  n <- length(x)  
  if(is.character(densfun)) {  
    ...  
    if(distname %in% c("lognormal", "log-normal")) {  
      if(!is.null(start))  
        stop(gettextf("supplying pars for the %s distribution is not supported",  
                      "log-Normal"), domain = NA)  
      if(any(x <= 0))  
        stop("need positive values to fit a log-Normal")  
      lx <- log(x)  
      sd0 <- sqrt((n-1)/n)*sd(lx)  
      mx <- mean(lx)  
      estimate <- c(mx, sd0)  
      sds <- c(sd0/sqrt(n), sd0/sqrt(2*n))  
      names(estimate) <- names(sds) <- c("meanlog", "sdlog")  
      vc <- matrix(c(sds[1]^2, 0, 0, sds[2]^2), ncol = 2,  
                  dimnames = list(names(sds), names(sds)))  
      names(estimate) <- names(sds) <- c("meanlog", "sdlog")  
      return(structure(list(estimate = estimate, sd = sds, vcov = vc, n = n,  
                          loglik = sum(dlnorm(x, mx, sd0, log=TRUE))),  
                    class = "fitdistr"))  
    }  
    ...  
  }  
  ...  
  structure(list(estimate = res$par, sd = sds, vcov = vc,  
                loglik = - res$value, n = n), class = "fitdistr")  
}
```

`mle()` allows to compute MLE by providing the minus log-likelihood and a guess estimate.

```
> library(stats4)
> y <- c(26, 17, 13, 12, 20, 5, 9, 8, 5, 4, 8)
> nLL <- function(lambda) -sum(stats::dpois(y, lambda, log = TRUE))
> mle(nLL, start = list(lambda = 5), nobs = NROW(y))
```

Call:

```
mle(minuslogl = nLL, start = list(lambda = 5), nobs = NROW(y))
```

Coefficients:

```
lambda
11.54545
```

`mle()` relies on `optim()` to carry out the numerical optimization and provides S4 object.

```
mle <- function(minuslogl, start, optim = stats::optim,
               method = if(!useLim) "BFGS" else "L-BFGS-B",
               fixed = list(), nobs, lower, upper, ...)
{
  ...
  oout <-
    if (length(start))
      if(!useLim)
        optim(start, f, method = method, hessian = TRUE, ...)
      else
        optim(start, f, method = method, hessian = TRUE, lower = lower, upper = upper, ...)
    else list(par = numeric(), value = f(start))
  coef <- oout$par
  vcov <- if(length(coef)) solve(oout$hessian) else matrix(numeric(), 0L, 0L)
  min <- oout$value
  fullcoef <- l2v(fullcoef) # to get the names right
  fullcoef[!isfixed] <- coef
  fullcoef[isfixed] <- fixed[isfixed]
  new("mle", call = call, coef = coef, fullcoef = fullcoef, fixed = fixed,
      vcov = vcov, min = min, details = oout, minuslogl = minuslogl,
      nobs = if(missing(nobs)) NA_integer_ else nobs,
      method = method)
}
```

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SCHEMATIC VIEW



SCHEMATIC VIEW



the **fitdistrplus** package allows

- different estimation methods: MLE, MME, QME, MGE, MSE
- different optimization algorithms,
- fixing a parameter to a given value,
- possibly left, right, interval censoring,
- bootstrap analysis.

18 YEARS OF DEVELOPMENT

Version 0.1-1

- `fitdist()` **calls** `mledist()`, `momdist()`
- **generic methods** : `plot`, `print`, `summary`



Version 1.2-6

- `fitdist()` **calls** `mledist()`, `mmedist()`, `qmedist()`, `mgedist()`, `msedist()`
- **generic methods** : `AIC`, `BIC`, `coef`, `logLik`, `plot`, `print`, `quantile`, `summary`, `vcov`
- `bootdist()` **calls** `mledist()`, `mmedist()`, `qmedist()`, `mgedist()`, `msedist()`

See [CRAN archive](#)

- `mledist()` allows a constrained optimization either by BFGS (via `optim()`) or by Nelder-Mead (via `constrOptim()`) of the normalized log-likelihood

$$\theta \mapsto \frac{1}{n} \sum_{i=1}^n \log f_X(x_i, \theta),$$

- initial iterate is computed by a closed-form estimator based on moments or quantiles, see `vignette(package="fitdistrplus", topic="starting-values")`.

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Examples

- gamma by moments : $\hat{\alpha} = m_2^2 / \mu_2, \hat{\theta} = \mu_2 / m_1$.
- Gumbel by quartiles : $\hat{\theta} = \frac{q_1 - q_3}{\log(\log(4/3)) - \log(\log(4))}, \hat{\alpha} = \hat{\theta} \log(\log(2)) + q_2$.

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GOODNESS OF FIT STATISTICS VIA `gofstat()`

- Simulate a gamma sample ($n = 100$)
- Fit exp. and gamma distributions to retrieve true param.

	MLE gamma	MLE exponential
Kolmogorov-Smirnov statistic	0.05	0.32
Cramer-von Mises statistic	0.04	3.50
Anderson-Darling statistic	0.27	17.59
Akaike's Information Criterion	862.36	974.23
Bayesian Information Criterion	867.57	976.84

TABLE: GoF stat. and criteria – Gamma, exponential

- Simulate a Poisson sample ($n = 100$)
- Fit Poisson and geom. distributions to retrieve true param.

	obscounts	theo MLE Poisson	theo MLE geometric
≤ 1	21.00	19.77	43.66
≤ 2	23.00	22.33	14.05
≤ 3	16.00	22.40	10.55
≤ 4	23.00	16.86	7.92
≤ 5	9.00	10.15	5.94
> 5	8.00	8.49	17.89

TABLE: Chi'square table – Poisson, geometric

	MLE Poisson	MLE geometric
Akaike's Information Criterion	391.37	452.44
Bayesian Information Criterion	393.98	455.05

TABLE: GoF statistics – Poisson, geometric

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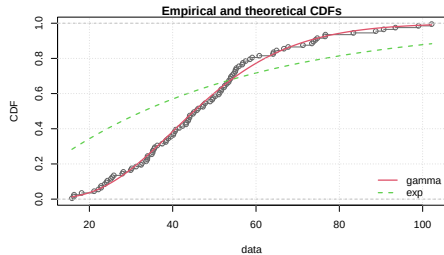
- Quantitative
- Graphics

5 EXTENSIONS AND INHERITANCE

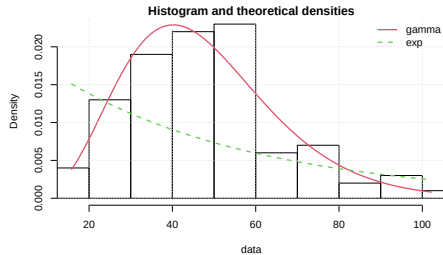
- the mbbefd package
- the OneStep package

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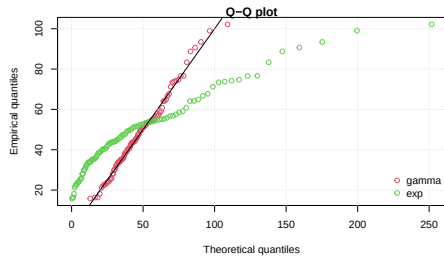
7 CONCLUSION



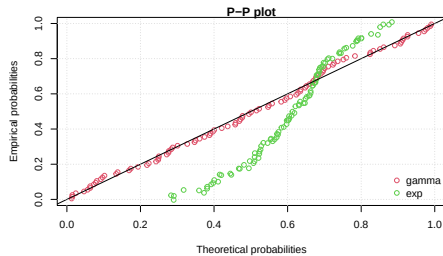
(A) produced by `cdfcomp()`



(B) produced by `denscomp()`



(C) produced by `qqcomp()`



(D) produced by `ppcomp()`

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EXPOSURE CURVES

The exposure curve function of X is the following increasing and concave function

$$G_X(d) = \frac{E(\min(X, d))}{E(X)}, \quad d \in [0, 1].$$

■ direct link between the c.d.f. and the exposure curve

$$G_X(d) = \frac{\int_0^d (1 - F_X(x)) dx}{\int_0^1 (1 - F_X(x)) dx} \Leftrightarrow F_X(x) = \left(1 - \frac{G'_X(x)}{G'_X(0)}\right) \mathbb{1}_{[0,1[}(x) + \mathbb{1}_{[1,+\infty[}(x).$$

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Examples

- uniform $]0, 1[$: $G_X(d) = d(2 - d)$.
- Dirac at 1 : $G_X(d) = d$.

Introduced by [1], the MBBEFD distribution is defined by the following exposure curve for $(a, b) \in \mathcal{D}_{a,b}$

$$\forall x \in I, G_X(x) = \begin{cases} \frac{\ln(\frac{a+b^x}{a+1})}{\ln(\frac{a+b}{a+1})} & \text{if } a(1-b) > 0 \\ \frac{1-b^x}{1-b} & \text{if } a = +\infty \text{ and } b < 1 \\ x & \text{if } a = 0 \text{ or } b = 1. \end{cases} \quad (1)$$

where the parameter domain is $\mathcal{D}_{a,b} = \{(a, b), a+1 > 0, a(1-b) \geq 0, b > 0\}$.

NB: *MBBEFD* stands for Maxwell Boltzmann Bose Einstein Fermi Dirac.

INHERANCY

fitdistrplus

CLASS FITDIST

- print, summary
- plot
- quantile
- logLik, AIC, BIC
- coef, vcov

CLASS BOOTDIST

- print, summary
- plot, density
- quantile



mbbefd

CLASS FITDR

no new function defined



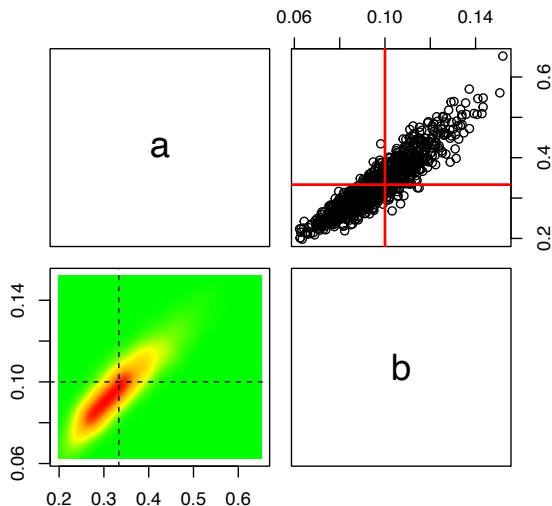
CLASS BOOTDR

only `vcov` is defined

SIMULATION ANALYSIS FOR A MBBEFD

- simulate with $a = 1/3$, $b = 1/10$
- fit and make bootstrap analysis
- sample size $n = 1000$, replicate $B = 1000$

Bootstrapped values of parameters



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ONE-STEP METHOD FROM [2]

As soon as the Fisher matrix is sufficiently regular, Le Cam's one-step estimation procedures can be achieved

$$\bar{\theta}_n = \theta_n^* + \tilde{\mathcal{I}}(\theta_n^*)^{-1} \cdot \frac{1}{n} \sum_{j=1}^n \dot{\ell}(\theta_n^*, X_j), \quad \tilde{\mathcal{I}}(\theta_n^*) = \begin{cases} -E(\ddot{\ell}(\theta)) \\ -\frac{1}{n} \sum_{j=1}^n \ddot{\ell}(\theta, X_j) \end{cases} \quad (2)$$

where $(\theta_n^*, n \geq 1)$ is the initial guess estimators, $\dot{\ell}$ is the gradient, $\ddot{\ell}$ is the Hessian matrix.

THEOREM

If θ_n^ is a $\sqrt{n}$ -consistent estimator of θ , then $\bar{\theta}_n$ is asymptotically normal and efficient (compared to MLE).*

INHERANCY – ONESTEP

fitdistrplus

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- print, summary
- plot
- quantile
- logLik, AIC, BIC
- coef, vcov

CLASS BOOTDIST

- print, summary
- plot, density
- quantile

OneStep

CLASS ONESTEP

no new function defined



NO NEW CLASS FOR BOOTSTRAP
nothing

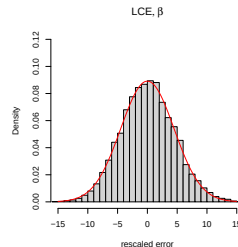
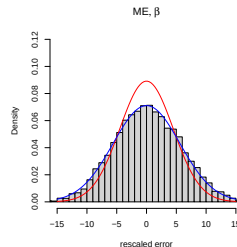
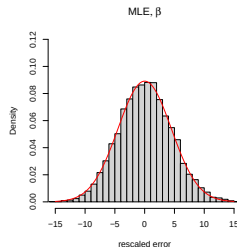
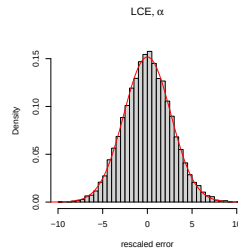
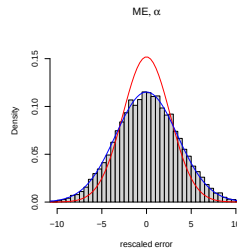
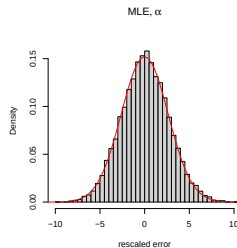
EXAMPLE WITH A GAMMA DISTRIBUTION

- two parameters :
shape α , rate β
- explicit Fisher
information matrix
$$\mathcal{I}(\theta) = \begin{pmatrix} \psi^{(2)}(\alpha) & -\frac{1}{\beta} \\ -\frac{1}{\beta} & \frac{\alpha}{\beta} \end{pmatrix}.$$

- example : sample size
 $n = 10000$, replicate
 $B = 10000$

- time (sec)

MLE	ME	LCE
1559.74	29.72	37.46



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HANDLING PROBABILITY DISTRIBUTIONS IN JULIA

The **Distributions** package provides a large collection of probabilistic distributions and related functions.

- Sampling from distributions
- Moments (e.g mean, variance, skewness, and kurtosis), entropy, and other properties
- Probability density/mass functions (pdf) and their logarithm (logpdf)
- Moment-generating functions and characteristic functions
- Maximum likelihood estimation
- Distribution composition and derived distributions (e.g., truncated distributions, censored distributions)

SAMPLING AND FITTING MLE

```
using Distributions
```

```
using Plots
```

```
mydist = Beta(2, 3)
```

```
x = range(0, 1.0, length=100)
```

```
mydensity = Distributions.pdf(mydist, x)
```

```
plot(x, mydensity)
```

```
y = Distributions.rand(mydist, 1000)
```

```
histogram(y, bins=30, normalize=:pdf)
```

```
fit(Beta, y)
```

```
sum(logpdf.(Beta, y))
```

SAMPLING AND FITTING MLE

```
using Distributions
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```

```
fit(Beta, y)
```

```
sum(logpdf.(Beta, y))
```

- no genericity of probability distribution (only a preselected set)
- no package for QME, MME, MSE, except **DistributionFits** for 6 distributions
- graphical comparison to be done manually
- GoF statistics to be done manually
- bootstrap should be carried out with **Bootstrap**

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CONCLUSION

fitdistrplus has advantages

- a complete workflow to fit and select distributions for any $d, p, q, r < \text{distribution}>$
- a variety of diagnostic tools, GoF statistics, likelihood profiling
- many estimation methods, in particular for interval-censored data
- bootstrap analysis (parallelized)
- a detailed [FAQ](#)

CONCLUSION

fitdistrplus has advantages

- a complete workflow to fit and select distributions for any `d, p, q, r <distribution>`
- a variety of diagnostic tools, GoF statistics, likelihood profiling
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Competitors in R are

- **distrMod** allows for a flexible MLE and MGE using S4 class.
- **ExtDist** provides a framework for an extensive set of probability distributions.
- **gamlss** fits the Generalized Additive Models for Location Scale and Shape (non i.i.d. setting)

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- **distrMod** allows for a flexible MLE and MGE using S4 class.
- **ExtDist** provides a framework for an extensive set of probability distributions.
- **gamlss** fits the Generalized Additive Models for Location Scale and Shape (non i.i.d. setting)

Distributions.jl is a good starting point in Julia but many tools are missing.

REFERENCES



S. Bernegger (1997),
The Swiss Re Exposure Curves and the MBBEFD
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OUTLINES

8 MSE

9 EXAMPLES

10 DESTRUCTION RATES

MAXIMUM SPACING ESTIMATION (MSE)

Consider a sample of observations $(x_1, \dots, x_n)$ (assuming real-valued observations).

- We can compute order statistics as $x_{(1)} < \dots < x_{(j)} < \dots < x_{(n)}$.

- Spacings on the distribution function $F(\cdot; \theta)$ are defined as

$$D_i(\theta) = F(x_{(i)}; \theta) - F(x_{(i-1)}; \theta), \quad i = 1, \dots, n+1$$

where $x_{(0)} = -\infty$ and $x_{(n+1)} = +\infty$.

- we know $D_i(\theta) > 0$ for any continuous cdf.

MSE consists in maximizing the average of the spacing logarithm

$$S_n(\theta) = \frac{1}{n+1} \sum_{i=1}^{n+1} \log D_i(\theta).$$

Under certain regularity conditions, MSE has asymptotically a normal distribution as MLE, see Ekstrom (2015).

OUTLINES

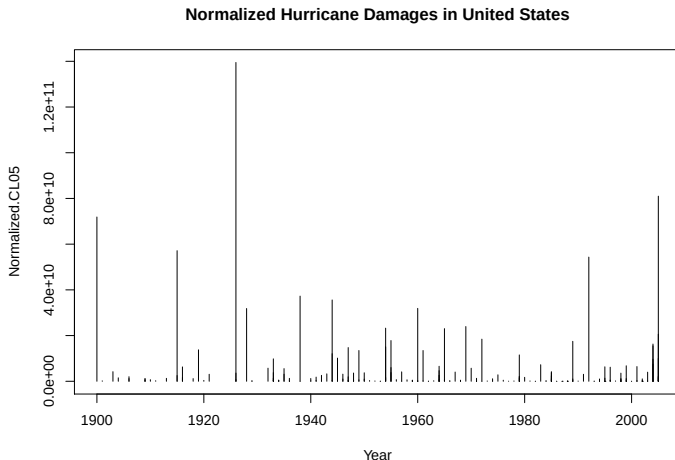
8 MSE

9 EXAMPLES

10 DESTRUCTION RATES

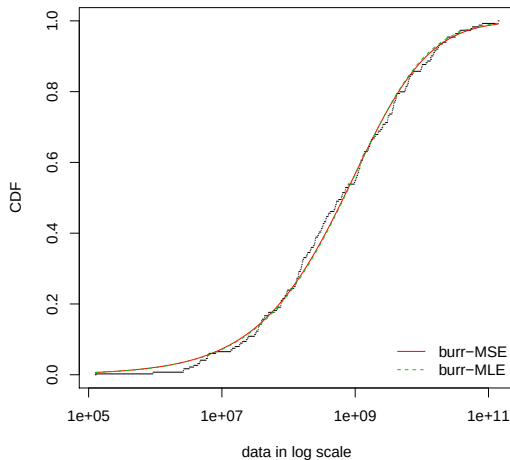
EXAMPLE 3 – REAL DATASET

Consider Normalized Hurricane Damages in the United States: 1900-2005 used in Pielke et al. (2008).

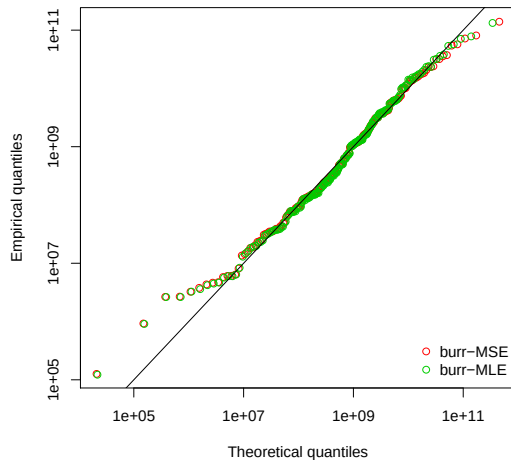


EXAMPLE 3 – REAL DATASET – GRAPHICS

Empirical and theoretical CDFs



Q-Q plot



EXAMPLE 3 – REAL DATASET – GOF. STATISTICS FOR BURR DISTRIBUTION

Coefficients values		
	MSE	MLE
shape1	2.2075	2.4563
shape2	0.5611	0.5604
rate	2.5318e-10	1.9825e-10

Goodness-of-fit statistics		
	MSE	MLE
Kolmogorov-Smirnov statistic	0.04421733	0.04867166
Cramer-von Mises statistic	0.06798511	0.08520099
Anderson-Darling statistic	0.43766768	0.51206510

Goodness-of-fit criteria		
	MSE	MLE
Akaike's Information Criterion	9301.882	9301.723
Bayesian Information Criterion	9311.880	9311.721

OUTLINES

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The distribution function is

$$\forall x \in I, F_X(x) = \begin{cases} \left(1 - \frac{(a+1)b^x}{a+b^x}\right) \mathbb{1}_{[0,1[}(x) + \mathbb{1}_{[1,+\infty[}(x) & \text{if } a(1-b) > 0 \\ (1 - b^x) \mathbb{1}_{[0,1[}(x) + \mathbb{1}_{[1,+\infty[}(x) & \text{if } a = +\infty \text{ and } b < 1 \\ \mathbb{1}_{[1,+\infty[}(x) & \text{if } a = 0 \text{ or } b = 1. \end{cases}$$

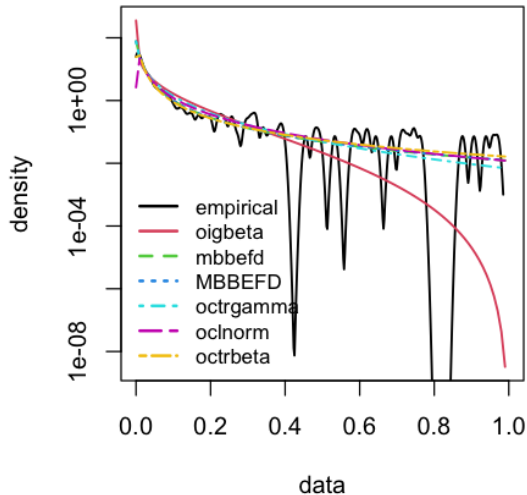
A mixed-type distribution with

- a probability mass $P(X = 1) = \frac{(a+1)b}{a+b}$,
- an improper density

$$\tilde{f}_X(x) \begin{cases} -\frac{a(a+1)b^x \ln(b)}{(a+b^x)^2} \mathbb{1}_{[0,1[}(x) & \text{if } a(1-b) > 0 \\ -\ln(b)b^x \mathbb{1}_{[0,1[}(x) & \text{if } a = +\infty \text{ and } b < 1 \\ 0 & \text{if } a = 0 \text{ or } b = 1. \end{cases}$$

CASE STUDY – FRENCH MARINE DATASET

Emp./theo. densities



Emp. and theo. exposure curve(s)

